

Edoardo Mosca

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XAI & NLP Researcher with a strong background in Mathematics, experienced in industry and academia. Holds a PhD in Computer Science and co-founded a startup. Author of 10+ scientific publications, led over 20 ML projects, and awarded from the Germany Ministry for research.



EXPERIENCE

Jan 2022 - Current

NLP Project Lead

Software Campus Program, Munich, Germany

I was selected by the German Ministry for research, who awarded me 100.000€ to lead and develop a project under the mentorship of a major German industry partner: the Holtzbrinck Publishing Group. Led a team of ~10 people to develop a platform for the interactive and controllable deployment of NLP models.

Aug 2022 - Current

XAI & NLP Academic Researcher

Technical University of Munich, Munich, Germany

In conjunction with my Doctoral degree, with a strong focus on post-hoc explainability to improve model interpretability, robustness, and human controllability. During this time I led over 20 ML projects, resulting in 11 scientific publications (@ ACL, COLING, MDPI, ECML, etc.).

Teaching (Master Level): Natural Language Processing | Advanced Natural Language Processing | Practical Courses - NLP Applications, XAI for Machine Learning.

Sep 2022 - Aug 2023

Co-Founder and CTO

Cryptoscape.ai, Berlin, Germany

I led the development of an NLP-powered platform for novice investors to quickly grasp blockchain projects and concepts. Our system simplifies complex information, promoting blockchain adoption. Our work has been supported from TU Berlin's and the KIEZ accelerator.

Tech Stack: Pytorch, FastAPI, Supabase.

Nov 2018 - Apr 2020

ML Research And Development Assistant

Infineon Technologies, Munich, Germany (*part-time*)

I applied computer vision and non-standard representation learning to estimate and optimize SoC hardware design metrics. My work resulted in three scientific publications where I co-authored as the 1st or 2nd author.

Tech Stack: Tensorflow, Keras, Python, bash.

Feb 2018 - Oct 2018

Data Scientist and Software Developer

ParkHere, Munich, Germany (*part-time*)

Designed, developed, and tested data-driven algorithms for automating pedestrian and car detection in large commercial parking lots, using data from multiple sensors tailored to parking slot types. My output of my work has been deployed in production.

Tech Stack: Python (Numpy, pandas, scikit-learn), Git, C++, SQL.



PROGRAMMING SKILLS

Data Science: Python  (+ Pandas, Numpy, Scikit-learn, Jupyter, etc..) | Mojo 

ML/NLP: Pytorch  | Hugging Face  | LangChain  | LMQL | Tensorflow-Keras 

Backend/BaaS: FastAPI  | Supabase 

Databases: PostgreSQL  (+ pgVector) | MongoDB 

OS / Version Control: Linux (bash)  | Git  **Other Languages:**  |  | Haskell 



EDUCATION

Aug 2020 – Sep 2023

Doctor in NLP and Explainable AI, Computer Science

Technical University of Munich, Germany

Research in the fields of Explainable AI and NLP. More specifically, my PhD thesis explores the usage of post-hoc explainability to work towards more interpretable, robust, and (human) controllable NLP models.

Dissertation: *Explainable AI for the Human-Centric Development of NLP Models*. **Supervisor:** Prof. Dr. Georg Groh.

Oct 2017 – Apr 2020

M.Sc. Mathematics in Data Science

Technical University of Munich, Germany

Strong focus on machine learning from both the mathematical (theoretical) and computer science (hands-on) perspectives. Several practical experiences in both computer vision and natural language processing. **Thesis:** *Explainability of Hate Speech Detection Models*.

Sep 2013 – May 2017

B.Sc. Mathematics

University of Pisa, Italy

Minor in computer science. **Thesis:** *Computational Classification for Regular Expression Membership Testing*



SELECTED PUBLICATIONS

ACL
2022

Edoardo Mosca et al. *"That Is a Suspicious Reaction!": Interpreting Logits Variation to Detect NLP Adversarial Attacks.*

COLING
2022

Edoardo Mosca et al. *SHAP-Based Explanation Methods: A Review for NLP Interpretability.*

AACL
2023

Edoardo Mosca et al. *IFAN: An Explainability-Focused Interaction Framework for Humans and NLP Models*

MDPI
2023

Edoardo Mosca et al. *Distinguishing Fact from Fiction: A Benchmark Dataset for Identifying Machine-Generated Scientific Papers in the LLM Era. (TrustNLP Extension)*



CERTIFIED TRAININGS

◦ **Agile for Management**
IAV, Berlin, 2 days, Aug 2022

◦ **Customer-Centric Design Thinking**
Grundfos, Denmark, 2 days, Sep 2022

◦ **Workplace Personalities**
Merck, Darmstadt, 2 days, Nov 2022

◦ **Hiring and Talent Acquisition**
Zeiss, Oberkochen, 2 days, Nov 2022

◦ **Change Management**
Trumpf, Stuttgart, 2 days, May 2023

◦ **Culture & Sustainability**
Huawei, Munich, 1 day, June 2023



LANGUAGES

Italian (Native) | English (C2) | German (C1) | Spanish (C1) | French (A2)



REFERENCES

Prof. Dr. Georg Groh (TUM)
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Dr. Lorenzo Servadei (Sony AI)
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Dr. Robert Seidl (Stats Perform)
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